When to request a hardware upgrade

Article KB0010035 | Owner: Desktop Support | Subcategory: Laptop slow / freezing | Applies to: Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840, MacBook Pro/Air (macOS 14/15), Windows 11 23H2/24H2, Managed via Microsoft Intune

Summary

This article helps users and Desktop Support agents decide when a laptop's sluggishness or freezing justifies a hardware upgrade (e.g., adding RAM or switching to SSD) versus performing software cleanup. It outlines typical symptoms, common causes, quick self-checks, a user-friendly optimization procedure, and the agent workflow for diagnosing upgrade needs, verifying results, and documenting the outcome.

Symptoms

- Laptop feels very slow or laggy
- System hangs or freezes during video calls or multitasking
- High CPU usage (often >90%) shown in Task Manager
- Applications become unresponsive or take a long time to launch
- Frequent disk activity light flashing even when idle

Common causes

- Insufficient RAM (typically <8?GB) for current workload
- Aging hard-disk drive (HDD) versus solid-state drive (SSD)
- Too many startup programs or background services
- Pending Windows updates that have not been applied for weeks
- Low free disk space (<10?% of drive)
- Malware or unwanted software consuming resources

Initial Checks

Before requesting a hardware upgrade, perform a few quick checks to rule out simple software issues. Verify that you are connected to the ACME-Corp Wi-Fi (machine certificate) or VPN if remote, and that you have logged into the Company Portal to see if any mandatory Intune policies are pending. Open Task Manager (Ctrl+Shift+Esc) and note the CPU, Memory, and Disk columns; if memory usage is consistently above 80?% or disk usage spikes to 100?% during normal work, those are strong indicators of a hardware bottleneck.

Self-Help Optimization Procedure

Follow these steps to improve performance without opening a ticket. Completing them may resolve the issue or provide useful data for the Desktop Support team if an upgrade is still needed.

1. 1. Save all work and close unnecessary applications.
2. 2. Open Task Manager -> Startup tab and disable any non-essential items (e.g., third-party updaters, chat apps).
3. 3. Run Disk Cleanup: search for "Disk Cleanup", select the C: drive, check Temporary files, Recycle Bin, and thumbnails, then click OK -> Delete Files.
4. 4. Ensure Windows is up to date: Settings -> Windows Update -> Check for updates. Install any available updates and restart when prompted.
5. 5. Open the Company Portal app, go to Devices -> your laptop -> Sync to force Intune policy refresh, then check for any pending application installations or updates.
6. 6. After completing the above, wait 10-15?minutes and reopen Task Manager. If memory usage drops below 70?% and the system feels responsive, the issue may be software-related. If it remains high, proceed to open a ticket with the gathered screenshots.

Service Desk Diagnostic Procedure

Agents use this workflow to confirm whether a hardware upgrade is warranted. Collect objective data, verify software health, and then make a recommendation based on Acme's hardware standards.

1. 1. Verify the ticket includes the user's laptop model, asset tag, and OS version (from Intune or Sysinfo).

2. 2. Ask the user to reproduce the issue while sharing their screen or providing a screenshot of Task Manager showing Memory and CPU usage.
3. 3. Confirm that:
   - Windows is fully patched (last Patch Tuesday <=5 days ago).
   - Disk free space >15?%.
   - No malware alerts in Defender for Endpoint.
   - Startup items are limited to Acme-approved applications.
4. 4. If memory usage stays >=80?% during normal productivity tasks (e.g., Teams call + Outlook + Chrome) and CPU usage frequently exceeds 85?%, note that the device likely lacks sufficient RAM.
5. 5. Check the current RAM configuration in Intune -> Devices -> device details -> RAM. Compare against the baseline for the model (most laptops ship with 8?GB; upgrade path is to 16?GB).
6. 6. If the device has an HDD, run a quick health check via Defender for Endpoint -> Device performance -> Disk health; if the HDD shows elevated latency or pending sectors, recommend SSD migration.
7. 7. Draft a recommendation:
   - Add RAM to 16?GB if slots are available and usage justifies it.
   - Replace HDD with SSD (or enable Intune-managed BitLocker-compatible SSD) if disk is the bottleneck.
   - If both RAM and storage are adequate, suggest further software investigation (e.g., profile cleanup, application reinstall).
8. 8. Communicate the recommendation to the user, obtain approval if required (per local procurement policy), and create a hardware change request in the IT portal.

Verification After Changes

After the Desktop Support team performs the upgrade (RAM addition, SSD swap, or both), verify that the improvement meets the user's expectations and that the device remains compliant with Acme policies.

1. 1. Ask the user to log in and run their typical workload (e.g., Teams meeting, Outlook, Chrome with several tabs).
2. 2. Have them capture a new Task Manager screenshot showing Memory and CPU usage.
3. 3. Confirm that Memory usage now stays below 60?% under load and that Disk active time is <20?% for an SSD.
4. 4. Ensure BitLocker (Windows) or FileVault (Mac) is still enabled and the recovery key is present in Intune.
5. 5. Run a quick Windows Update check to verify the device is still receiving patches.
6. 6. Close the ticket with a resolution note detailing the parts installed, service tag, and before/after performance metrics.

Documentation and Closure

Proper documentation helps track hardware life-cycle trends and supports future budgeting. Include all relevant details in the ticket and update the asset record if necessary.

1. 1. Record the original symptom, the diagnostic data collected (screenshots, RAM/disk specs), and the exact upgrade performed (part number, quantity).
2. 2. Note the time taken for the upgrade and any user-reported improvement.
3. 3. Update the asset's hardware profile in Intune or the CMDB to reflect the new RAM/storage capacity.
4. 4. Add a knowledge-base link to this article for future reference.
5. 5. Set the ticket status to Resolved and notify the user via email or Teams.

When to escalate

If the laptop is still under warranty and requires component replacement (RAM or SSD), escalate to Desktop Support with the asset tag, usage logs, and screenshots; the Desktop Support team will coordinate the warranty claim. For out-of-warranty devices, advise the user to submit a formal hardware upgrade request through the IT service portal (https://it.acme.example) including the justification gathered in the ticket. In cases where the upgrade requires specialized parts not stocked locally (e.g., proprietary DDR5 modules), escalate to the Desktop Support Lead who will engage the vendor procurement team.

Keywords: hardware upgrade, laptop slow, freezing, high CPU, RAM upgrade, SSD replacement, performance troubleshooting, Desktop Support, Intune, BitLocker, FileVator